REQUIREMENTS GATHERING

1. Stakeholder Analysis

Stakeholder	Needs and Expectations
Deaf and Hard-of-Hearing Users	An AI-Powered mobile application that recognizes sign language gestures in real time and displays the corresponding Arabic words.
People Who Do Not Understand Sign Language	A simple way to understand sign language gestures through Arabic text.
AI/ML Developer	Suitable sign language datasets, accurate gesture recognition, and successful integration of the AI model with the application.
Mobile Application Developer	A responsive and user-friendly mobile application integrated with the AI model through an online API.
UI/UX Designers	A simple, intuitive, and accessible interface that makes live translation easy to use.

2. User Stories & Use Cases

2.1 User Stories

- As a **deaf user**, I want to open the camera and sign **so that** my signs are converted to Arabic text in real time.
- As a **deaf user**, I want the app to show the recognized word immediately **so that** I can communicate in a live conversation.
- As a **hearing person**, I want to read the translated text on the screen **so that** I can understand the person signing.
- As a **user**, I want clear feedback when the sign isn't recognized **so that** I can repeat it.
- As a **user**, I want simple Arabic UI **so that** I can use the app without any training.

2.2 Use Cases

UC-01: Live Sign Recognition

Field	Description
Actor	Deaf user
Precondition	The app is open
Trigger	The user opens the camera screen

Main Flow:

1. The user opens the camera screen.
2. The user starts performing a sign in front of the camera.
3. The app sends the captured frames to the AI model.
4. The model recognizes the sign.
5. The corresponding Arabic word is displayed on the screen in real time.

Alternative Flow:

- **Low confidence:** If the model's confidence is below the threshold, the app displays the message "Please repeat the sign."

Postcondition: The recognized Arabic word is displayed on the screen.

UC-02: Grant Camera Permission

Field	Description
Actor	User
Precondition	The app is launched
Trigger	The user opens the live recognition feature

Main Flow:

1. The user grants permission.
2. The app starts live sign recognition.

Alternative Flow:

- **Permission denied:** The app displays a message explaining that camera access is required for the app to function.

Postcondition: Camera access is enabled and live recognition is active.

UC-03: Handle Recognition Failure

Field	Description
Actor	System
Precondition	Live recognition is active
Trigger	The system detects poor lighting or an unclear/undetected hand

Main Flow:

1. The system detects that the input quality is insufficient (e.g., low lighting or hand not clearly visible).
2. The system notifies the user to improve the conditions (e.g., adjust lighting or reposition the hand within the frame).
3. The user adjusts their position and continues signing.

Postcondition: The user is guided to restore proper recognition conditions, and recognition resumes.

3. Functional Requirements

ID	Functional Requirement
FR-01	The system shall provide a mobile application interface for accessing its main features.
FR-02	The system shall capture sign language gestures in real time.
FR-03	The system shall process the captured input using the sign language recognition AI model.
FR-04	The system shall communicate with the AI recognition service through an API or server, according to the implemented architecture.
FR-05	The AI model shall recognize supported Arabic sign language gestures and predict their corresponding words.
FR-06	The system shall display recognized words in Arabic on the mobile application's screen.
FR-07	The system shall continue live recognition while the translation feature is active.
FR-08	The system shall handle unrecognized gestures, network failures, and service errors appropriately.
FR-09	The system shall provide a Settings screen for the preferences supported by the application.

4. Non-functional Requirements

Category	Non-functional Requirement
Performance	The system should recognize gestures and display results with minimal delay to support real-time interaction.
Accuracy	The AI model should achieve an acceptable recognition accuracy for the supported sign language vocabulary, measured through testing.
Usability	The application should provide a clear, simple, and easy-to-navigate user interface.
Accessibility	Arabic text output should be readable, clearly presented, and easy to understand.
Security	Communication between the mobile application and the online recognition service should use HTTPS.
Privacy	Camera input and related data should be handled responsibly and should not be stored unnecessarily.
Reliability	The application should handle recognition errors, network interruptions, and API failures without unexpected crashes.
Compatibility	The application should operate on the Android devices and operating system versions supported by the project.
Maintainability	The mobile application, API, and AI model should be structured to facilitate testing, maintenance, and future improvements.
Availability	The online recognition service should be accessible when the service is operational and the user has an internet connection.

5. Project Scope and Limitations

In Scope

- Real-time recognition of supported Arabic sign language gestures.
- Translation from sign language gestures into Arabic written words.
- Live camera input through the mobile application.
- Integration between the mobile application and the AI recognition service.
- Displaying recognition results and handling errors.

Out of Scope

- Translating Arabic written text into sign language.
- Supporting languages other than Arabic.
- Saving and viewing translation history.
- Translating complete sentences unless this functionality is explicitly implemented and supported by the AI model.